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To the UTRIP Admissions Committee,
University of Tokyo Faculty of Science,

I write with genuine enthusiasm to recommend Kavish Kumar for the University of Tokyo Research Internship Programme. Kavish completed his final-year capstone project under my supervision during January to May 2025, and I consider him one of the more technically self-directed students I have mentored at Plaksha.

The project -- automatic speech recognition for dysarthric speakers -- is a technically demanding problem that most final-year undergraduates would have simplified considerably. Kavish did not. He chose to build a speaker-independent system, which requires the model to generalise across individuals whose speech varies significantly not just person-to-person, but day-to-day. He identified wav2vec 2.0 as an appropriate feature extractor after surveying the literature independently, implemented a bidirectional LSTM decoder, and handled the full pipeline: dataset alignment between LibriSpeech and TORGO, fine-tuning strategy, and evaluation. His final word error rate of 48% and character error rate of 27% are competitive with published speaker-dependent baselines -- a meaningful result for a six-month undergraduate project.

What distinguished Kavish was not just technical competence but research judgement. When an early fine-tuning run produced unexpectedly poor results, he did not simply re-run with different hyperparameters. He went back to the embeddings, visualised the latent space, and identified that the wav2vec representations for dysarthric speech were clustering differently than expected -- a finding that motivated a different fine-tuning schedule. That kind of diagnostic thinking is exactly what separates a researcher from a practitioner.

Beyond this project, I am aware of Kavish's statistical work on flight delay data and his experience at NTU Singapore. His range -- from signal processing to tabular ML to collaborative research in a foreign institution -- gives me confidence that he will adapt quickly to a new lab environment and contribute meaningfully within UTRIP's six-week window.

I recommend Kavish without reservation. He is technically prepared, intellectually curious, and -- from what I observed over five months of close supervision -- genuinely motivated by the questions rather than the credential. Please feel free to contact me directly if you would like to discuss his candidacy further.

Yours sincerely,

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